

Title: Project Registration & Progress Review**FF No. 180**

Department: Computer Engineering	Academic Year: 2026-27
Semester: I	Group No.: TY-I20
Project Title: Offline Agentic AI for Financial Fraud Detection and Investigation	
Project Area: FinTech, Artificial Intelligence, Machine Learning	

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Project approved / Not approved

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Project Synopsis

Abstract:

The rapid growth of digital banking and instant payment systems has made financial fraud increasingly sophisticated, exposing the limitations of traditional rule-based fraud detection and Anti-Money Laundering (AML) systems, which often generate excessive false alerts and require significant manual effort. This project proposes an **Offline Agentic AI platform for intelligent financial investigation** that operates entirely within an organization's infrastructure, ensuring data privacy, regulatory compliance, and independence from cloud-based AI services. The system combines machine learning, anomaly detection, graph-based relationship analysis, explainable AI, and a locally hosted large language model to identify suspicious transactions, analyze customer behavior, assess risk, uncover hidden financial networks, and generate clear investigation reports. A team of specialized AI agents collaborates to automate the investigation process while keeping human analysts involved in final decision-making, and an optional blockchain-based audit trail ensures transparency and accountability. The proposed solution aims to reduce false positives, accelerate investigations, improve explainability, and provide a practical offline framework for secure, intelligent, and trustworthy financial investigation in modern banking environments.

Introduction:

The rapid growth of digital banking, instant payment systems, and online financial services has significantly increased both the volume of transactions and the complexity of financial fraud. Modern cybercriminals use advanced techniques such as phishing, synthetic identities, deepfakes, and money-laundering networks, making traditional rule-based detection systems increasingly ineffective. Although artificial intelligence has improved fraud detection capabilities, many existing AI solutions lack transparency, making it difficult for banks to justify automated decisions and comply with regulatory requirements. At the same time, strict data privacy and sovereignty regulations have created a strong need for secure, on-premise AI solutions that do not rely on external cloud services. Agentic AI further enhances this capability by enabling multiple intelligent agents to collaborate on tasks such as analysing transactions, assessing customer risk, identifying suspicious relationships, and generating investigation reports, while keeping human experts involved in final decision-making. Together, these advancements highlight the need for an offline, explainable, and intelligent financial investigation platform that improves detection accuracy, reduces false alerts, streamlines investigations, and ensures regulatory compliance.

Problem Statement:

Current financial fraud detection and Anti-Money Laundering (AML) systems rely heavily on static rule-based approaches that generate a high volume of false alerts, require extensive manual investigation, and struggle to detect evolving fraud patterns and complex transaction networks. Existing AI-based solutions often lack explainability, making their decisions difficult to interpret and justify for regulatory compliance, while cloud-based AI introduces concerns related to data privacy, security, and data sovereignty. As a result, financial institutions face delayed investigations, fragmented workflows, and increased operational costs. Therefore, there is a need for a secure, offline, and explainable intelligent investigation platform that integrates machine learning, anomaly detection, graph analytics, and Agentic AI to assist investigators in identifying suspicious financial activities, prioritizing risks, generating transparent investigation reports, and supporting faster, more accurate, and compliant decision-making without exposing sensitive financial data to external systems.

Literature Review:

Sr. No	Authors	Yr	Methodology	Advantages	Limitations	Research Gap
1	Ali, R.	2026	XGBoost + regularised DNN with SHAP-based interpretability for credit-card fraud (Explainable AI Framework for Credit Card Fraud Detection)	High PR-AUC (0.98+) and MCC; global & local SHAP explanations; regulation-aligned	Works only on structured tabular data; cloud/offline deployment not addressed; single dataset	No offline deployment or multi-agent orchestration
2	Ojo, I. P.; Tomy, A.	2025	Random Forest / Gradient Boosting with SHAP & LIME on card-fraud data (Explainable AI for Credit Card Fraud Detection)	SMOTE handles imbalance; dual XAI (SHAP+LIME) improves stakeholder trust	Near-perfect accuracy suggests overfitting risk; no graph or behavioural features	No entity-relationship / AML-pattern modelling
3	Raufi, B.; Finnegan, C.; Longo, L.	2024	Comparative study of SHAP, LIME, ANCHORS and DICE on a dense neural network for card fraud (XAI'2024, Springer CCIS)	Rigorous comparison of interpretability methods on stability & fidelity	Focuses purely on explainability benchmarking, not detection or investigation workflow	No investigator-facing narrative generation
4	Hasan, M. R.; Gazi, M. S.; Gurung, N.	2024	Interpretable models for U.S. card-fraud compliance (Explainable AI in Credit Card Fraud Detection)	Ties explainability directly to regulatory trust and transparent decisioning	Limited to U.S. regulatory context; no AML or layering detection	Gap between XAI theory and AML-specific compliance reporting
5	Li, K. et al.	2024	SEFraud — graph-based self-explainable fraud detector using interpretative mask learning (ACM SIGKDD)	Learns both fraud prediction and explanation jointly on transaction graphs	Requires large labelled graph data; heavy GPU training cost	No lightweight offline / on-premise variant
6	Deprez, B.; Vanderschueren, T.; Baesens, B.; Verdonck, T.; Verbeke, W.	2024	Systematic review and experimental evaluation of network analytics for AML (arXiv:2405.19383)	Comprehensive benchmark of graph-based AML techniques across public datasets	Notes scarcity of real, labelled AML data; most methods evaluated only on synthetic/semi-synthetic graphs	Calls for reproducible, deployable AML pipelines — directly motivates this project

Sr. No	Authors	Yr	Methodology	Advantages	Limitations	Research Gap
7	Multiple authors (survey)	2025	Advances in Continual Graph Learning for Anti-Money Laundering Systems: A Comprehensive Review (arXiv:2503.24259)	Surveys continual/graph learning to counter concept drift in laundering patterns	Continual-learning AML research still scarce; catastrophic forgetting unresolved	No integration with agentic reasoning or human-in-the-loop investigation
8	Lo, W. W.; Kulatilleke, G. K.; Sarhan, M.; Layeghy, S.; Portmann, M.	2023	Inspection-L — self-supervised GNN node embeddings for Bitcoin money-laundering detection (Applied Intelligence)	Self-supervised approach reduces dependency on scarce labelled data	Evaluated only on Bitcoin/Elliptic-style ledgers; not generalised to fiat banking rails	No fusion with tabular / behavioural risk features
9	Lawal, O.; Okolie, A.; Obunadike, C.	2025	Explainable GNN framework for AML in cryptocurrency transactions using the Elliptic dataset (IJCNC)	Combines relational GNN learning with explainability for regulator-facing audit needs	Crypto-specific; placement–layering–integration stages not mapped to fiat banking typologies	Motivates a bank-rail-oriented graph module in the proposed system

Objectives:

1. Develop an offline, on-premise AI platform for financial-crime (fraud and AML) investigation that requires no external cloud AI service.
2. Detect fraudulent financial transactions using an ensemble of supervised machine-learning classifiers (XGBoost, LightGBM, Random Forest).
3. Detect AML typologies such as layering, structuring/smurfing, and mule-account networks using graph relationship analysis.
4. Deploy specialised Agentic AI agents to autonomously assist — not replace — human investigation workflows.
5. Generate explainable, SHAP-grounded fraud and AML reports for every flagged case.
6. Build a graph-based relationship-analysis engine capable of surfacing multi-hop, cross-account laundering chains.
7. Ensure customer transaction data never leaves the institution's own infrastructure.
8. Reduce false-positive alert volume relative to a static rule-based baseline.
9. Improve investigation speed through automated evidence gathering and narrative drafting.
10. Maintain immutable, tamper-evident audit records of every model decision and human override (optional permissioned blockchain).

Proposed system:

1. **Offline Agentic AI Platform** that operates entirely within an organization's infrastructure, ensuring data privacy, security, and regulatory compliance.
2. **Modular and layered architecture** enabling independent upgrades and maintenance of individual system components.
3. **Transaction Processing Module** to collect, normalize, and process financial transaction data from banking systems.
4. **Data Preprocessing & Feature Engineering** to clean data and extract behavioral, temporal, device, location, and transaction-based features.
5. **Fraud Detection Engine** using machine learning and anomaly detection algorithms to identify suspicious transactions.
6. **AML Detection Engine** to detect money laundering patterns such as structuring, layering, rapid fund transfers, and mule accounts.
7. **Customer Behavior Analysis Module** to identify unusual activities by comparing transactions with historical customer behavior.
8. **Graph Relationship Analysis** to discover hidden connections between accounts and identify suspicious transaction networks.
9. **Risk Scoring Module** to generate a unified risk score by combining fraud, AML, behavioral, and graph analysis results.
10. **Explainable AI (XAI) Module** to provide transparent, feature-level explanations for every prediction and risk assessment.
11. **Local LLM-based Investigation Agent** to analyze evidence and generate human-readable investigation summaries without using cloud services.
12. **Automated Report Generation** for creating regulator-ready investigation reports and Suspicious Activity Reports (SARs).
13. **Interactive Dashboard** for investigators to monitor alerts, visualize transaction networks, review risk scores, and manage cases.
14. **Audit Logging System** to maintain a secure record of all model predictions, AI actions, and investigator decisions.
15. **Human-in-the-Loop Workflow** ensuring investigators review and approve all critical AI-generated decisions before final action.

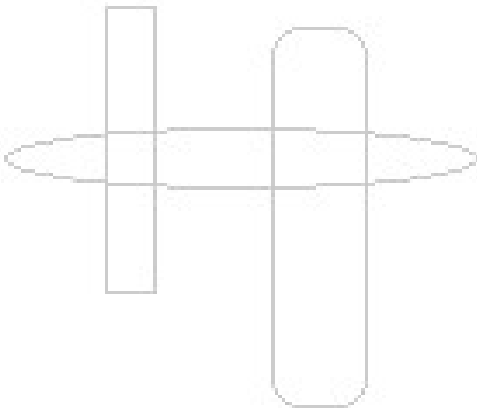
Tech Stack:

Layer	Technologies	Purpose
Frontend	React.js, Material UI, Plotly / Recharts	Investigator dashboard, case queues, graph visualisation
Backend / API	FastAPI (Python), Uvicorn/Gunicorn	REST + WebSocket services for scoring, agents, and case management
Core Language	Python 3.11	ML pipelines, agent orchestration, feature engineering
Machine Learning	Scikit-learn, XGBoost, LightGBM, PyTorch	Supervised fraud classifiers and autoencoder anomaly models

Layer	Technologies	Purpose
Local / Offline LLM	Ollama runtime with Llama 3, Qwen, or Mistral	On-premise natural-language investigation and report generation
Agent Orchestration	LangGraph / custom finite-state agent framework	Coordinates the eight specialised investigation agents
Database	PostgreSQL	Transaction store, case records, audit logs
Cache / Queue	Redis, Celery	Asynchronous scoring jobs and background agent tasks
Visualization	Plotly, Dash	Risk dashboards and graph-relationship views
Containerization	Docker, Docker Compose	Reproducible, air-gapped deployment
Version Control	Git, GitHub (private/self-hosted mirror for offline sites)	Source control and CI/CD
Optional Ledger	Hyperledger Fabric (permissioned)	Immutable, tamper-evident audit trail of decisions and approvals

Group No.	TY-I20		
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

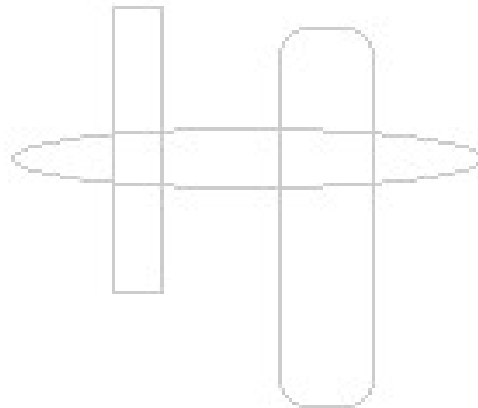
Review No.: 1	Group No.: TY-I20	Date:
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Review No.: 2

Group No.: TY-I20

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Progress Review Report



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